

3. All leading entries in a row have only zeros below them in their respective columns.

$$\text{Row 2, Column 1} = 0 \quad \text{Row 3, Column 1} = 0$$

$$\text{Row 3, Column 2} = 0$$

$\therefore$ The condition is satisfied and the matrix is in echelon form.

For the matrix to be in reduced echelon form, the following extra conditions must be satisfied:

4. The leading entry in each nonzero row is 1. We have seen this is true.

5. Each leading 1 is the only nonzero entry in its column.

We have already checked the column entries below a leading entry. Now we shall check the column entries above the leading entry.

$$\text{Row 1, Column 2} = 0$$

$$\text{Row 1, Column 3} = 0 \quad \text{Row 2, Column 3} = 0$$

$\therefore$ The condition is satisfied.

The matrix is in reduced echelon form $\square$

$$b) \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

1. Each of the rows has at least one nonzero element. $\therefore$ The condition that all nonzero rows are above all zero rows is vacuously true.

2. Row 1 Leading entry: 1 in Column 1

Row 2 Leading entry: 1 in Column 2

Row 4 Leading entry: 1 in Column 4

$\therefore$ The condition that the leading entry of a row must be in a column to the right of the columns of all leading entries of rows above it is valid.

$$\text{Row 2, Column 1} = \text{Row 3, Column 1} = 0$$

$$\text{Row 3, Column 2} = 0$$